
<Finding-first title: state the mechanism or the result, not the topic>

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Abstract

<TODO move 1: context and stakes of the simulated system, in one clause.> <TODO move 2: the specific gap, phrased as a mechanism, not as "few studies".> <TODO move 3: our model and the single design decision that drives the result.> <TODO move 4: headline number, with baseline and uncertainty.> <TODO move 5: the implication, stated plainly.>

Keywords: multi-agent simulation; <domain>; <method family>; <mechanism>; reproducibility

1 Introduction

<TODO P1: domain and stakes. One number. One citation, e.g. (Helbing and Molnár, 1995).>

<TODO P2: the specific problem, narrowed. Define the decision that has to be made.>

<TODO P3: why the obvious approach fails, by mechanism. Not "existing methods have limitations" but "routing on instantaneous queue length is self-defeating because the act of choosing a station changes the quantity being chosen on".>

<TODO P4: our approach, and the intuition that motivates it.>

Contributions

- C1. <A formal model of ... (discharged by Sec. 4)>
- C2. <A mechanism-level explanation of why ... (discharged by ??)>
- C3. <An allocation rule that ... (discharged by Table 2, Figure 1)>
- C4. <A cross-implementation parity check that ... (discharged by ??)>

2 Related Work

<TODO: 3-5 contrast paragraphs, each ending in a concrete difference.>

Table 1. Notation. Every symbol used in Sec. 4 must appear here, and no symbol that appears only once in the paper should.

Symbol	Meaning	Domain
N	number of agents	$\mathbb{N}$
S	set of service stations	—
c_j	capacity (service rate) of station j	$\mathbb{R}_{>0}$
T	makespan (last completion time)	$\mathbb{R}_{\geq 0}$
T^{lb}	capacity lower bound on T	$\mathbb{R}_{\geq 0}$

3 Problem Formulation

<TODO: system definition, agent set, state, action, observation, transition.>

<TODO: objective and constraints. Give the lower bound / best case if one exists — a bound turns every later number into a ratio and makes the results interpretable.>

Assumptions

1. <Statement.> *Justification:* <why it holds in this domain.> *If relaxed:* <which result changes and in which direction.>

4 Model and Method

4.1 Model

<TODO: transitions, decision epochs, information available to each agent, and the crucial question of WHEN an agent may re-decide.>

4.2 Allocation rule

<TODO: state the rule, then prove or argue the property it has.>

Proposition 1. <A property of the rule, stated so it can be falsified.>

Proof. <Proof, or, for an empirical claim, an explicit pointer to the experiment that tests it. Do not dress up a simulation observation as a theorem.> $\square$

Design rationale. <Why this rule and not the two obvious alternatives.>

4.3 Algorithm

<TODO: pseudocode only if the procedure cannot be stated in prose; a 40-line listing that restates the equations is padding.>

5 Experiments

5.1 Setup

<TODO: instance generation, size, seeds, hardware, wall-clock. State the number of independent runs and the seed policy — "we report the mean over 10 seeds" without a seed policy is not reproducible.>

5.2 Baselines

<TODO: one sentence per baseline, naming its class and its information access. A baseline given less information than the proposed method is not a baseline.>

Table 2. Main comparison on <instance family>. Mean $\pm$ s.d. over $n = k$ seeds. Bold marks the best mean within one standard deviation; T^{lb} is the capacity lower bound from Sec. 3.

Method	T (min)	T/T^{lb}	Class
Nearest station	483.94	3.387	rule
Random assignment	238.56	1.670	rule
Capacity-proportional (ours)	157.01	1.099	rule + bound
Capacity lower bound	142.86	1.000	—

5.3 Main result

<TODO: one paragraph that reads the table. State the effect size, the direction, and the mechanism that explains it. "Our method performs better" is not a reading of a table; "the gap closes from 3.4x to 1.1x of the lower bound, which means the remaining inefficiency is no longer allocation but travel" is.>

Takeaway

<One sentence the reader should be able to repeat from memory after closing the paper. If you cannot write this sentence, the experiment does not have a result yet.>

5.4 Mechanism

<TODO: the intermediate quantity that carries the effect.>

5.5 Ablation

<TODO: one row per design decision, each removed individually. An ablation that changes two things at once explains nothing.>

5.6 Sensitivity and robustness

<TODO: vary the parameters the conclusion depends on, and report the range over which the conclusion survives. "Robust" must come with the range in which it is robust.>

5.7 When it fails

<TODO: the regime where the method loses, with the numbers that show it. A paper with no failure case is either trivial or not yet understood.>

6 Discussion

<TODO: what the mechanism implies beyond this instance, and what would falsify the mechanism. Keep it short; speculation padded to a page is worse than one honest paragraph.>

7 Limitations

<TODO: three concrete limits with the direction of the bias each induces.>

Reproducibility Statement

Reproducibility.

<TODO: where the code, instances and seeds live; which command reproduces which figure; the exact software versions. A statement that says "code available on request" is worth nothing and `csf_readiness.py` reports it.>

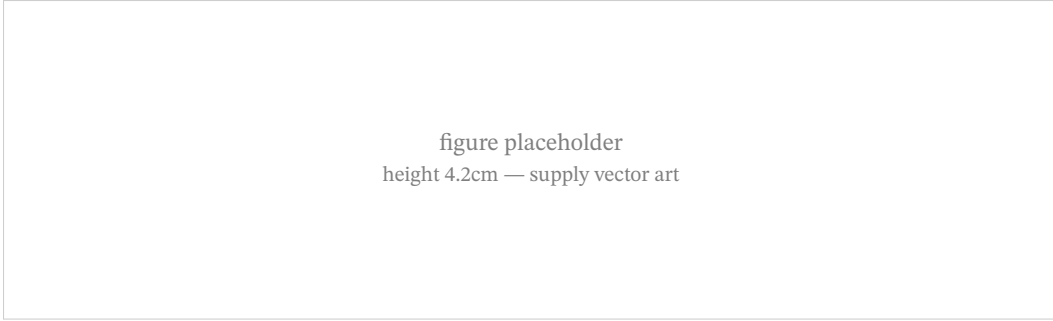


Figure 1. <What the reader must conclude, not what the figure contains.> Panels: (a) <role>; (b) <role>. Labels and vector art are produced separately; see `figures/contracts/` for the label ledger.

Compute Statement

Compute.

<TODO: hardware, wall-clock, number of runs. Required by ICML and increasingly by everyone else.>

References

Dirk Helbing and Péter Molnár. Social force model for pedestrian dynamics. *Physical Review E*, 51(5):4282–4286, 1995.